Uplift Modeling on the Hillstrom Email Dataset

Results Report — Causal Targeting Analysis with Meta-Learners

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2026 · Data: Kevin Hillstrom's MineThatData e-mail experiment ($n \approx 64{,}000$)

1 Executive Summary

This report summarizes a randomized-controlled-trial (RCT) uplift modeling analysis on the Hillstrom email dataset: 64,000 customers randomized into a *Men's E-Mail*, a *Women's E-Mail*, or a *No E-Mail* arm. The goal is not to predict who will respond, but to predict *who responds because of the campaign* — the individual treatment effect (ITE), estimated via S-/T-/X-Learner meta-learners on a *visit* outcome.

Headline result: a predicted-uplift targeting policy is statistically supported for the **Women's E-Mail** campaign (best model: X-Learner GBM, held-out Qini AUC = 0.076, 95% CI [0.042, 0.111], excludes zero) but **not** for the **Men's E-Mail** campaign (best point estimate 0.012, but every model's 95% CI includes zero — statistically indistinguishable from random targeting).

Every source surveyed reports Hillstrom as a small, volatile uplift signal relative to other benchmark datasets, regardless of the exact Qini convention used — consistent with this project's own null Men's-arm result. A specific published numeric range (0.019–0.035) is shown for order-of-magnitude context in Section 6, but is flagged there as *not confirmed* to use the same normalization as this project's figures.

2 Data and Causal Identification

Because Hillstrom is a genuine RCT (not observational data), a causal read of the estimated effects is licensed *if* randomization held up in practice. This was checked, not assumed:

- **Covariate balance:** standardized mean differences between each e-mail arm and control were uniformly small (max $|\text{SMD}| \approx 0.014$, well under the conventional 0.1 threshold) across recency, history, purchase-category flags, newbie status, zip region, and channel.
- **Design matrix:** one-hot encoding of `zip_code`/`channel` is rank-deficient without `drop_first` (9 cols, rank 7) and full rank with it; VIF values all < 5 (max ≈ 3.05).
- **Assumptions flagged:** ignorability holds by design (randomization); SUTVA is plausible for an e-mail campaign (no cross-customer interference).

The 3-arm design is reduced to **two parallel binary comparisons** (Men's vs. No E-Mail; Women's vs. No E-Mail) rather than pooling both e-mail arms, since the two campaigns are different interventions, not two doses of one treatment. Each two-arm subset of a randomized 3-arm design is itself a valid RCT. The two comparisons share the same control group, so their sampling noise is correlated — no direct cross-arm claim (e.g. "Men's uplift $>$ Women's uplift") is made anywhere in this report.

The outcome modeled is **visit** ($\approx$15% base rate), not **conversion** ($<$1%). A dedicated stress test (Women's arm, X-Learner) confirmed conversion is too sparse for reliable heterogeneous-effect estimation: 103 conversions out of 12,832 treated rows vs. 68 of 12,784 control rows in the training data — too few positive events, especially for T-/X-Learner's split-sample stages.

3 Modeling Approach

3.1 Meta-learners

Three meta-learner families were used, all with `HistGradientBoostingClassifier` as the base learner (tuned once via `RandomizedSearchCV` against cross-validated PR-AUC on the pooled visit outcome — a lower-variance proxy than tuning directly against the noisy, small-holdout Qini metric):

S-Learner — treatment as a feature of a single pooled model:

$$\hat{\mu}(x, t) \text{ fit jointly on } (X, T); \qquad \hat{\tau}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$$

T-Learner — one model per arm:

$$\hat{\mu}_1(x) \text{ fit on treated only, } \hat{\mu}_0(x) \text{ fit on control only;} \qquad \hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$$

X-Learner — imputed-effect regressors blended by the (known, randomization-derived) propensity $g(x)$:

$$\begin{aligned} D_i^1 &= Y_i - \hat{\mu}_0(X_i) \text{ for treated,} \qquad D_i^0 = \hat{\mu}_1(X_i) - Y_i \text{ for control} \\ \hat{\tau}_1(x) &= \mathbb{E}[D^1 \mid X{=}x], \qquad \hat{\tau}_0(x) = \mathbb{E}[D^0 \mid X{=}x] \\ \hat{\tau}(x) &= g(x) \cdot \hat{\tau}_0(x) + [1 - g(x)] \cdot \hat{\tau}_1(x) \end{aligned}$$

A logistic-regression S-Learner is carried as a linear baseline throughout. scikit-uplift (`SoloModel`/`TwoModels`) implements S-/T-Learner; X-Learner is a hand-rolled implementation, since scikit-uplift ships no native X-Learner.

3.2 Evaluation metrics

Qini AUC — sklift's *normalized* Qini coefficient (Radcliffe, 2007): the area between the model's cumulative incremental-gain curve and the random-targeting baseline, divided by that same area for the theoretical perfect-targeting curve:

$$\text{Qini}(k) = Y_t(k) - Y_c(k) \cdot \frac{N_t(k)}{N_c(k)} \qquad \text{[cumulative gain at fraction } k\text{]}$$

$$\text{Qini AUC} = \frac{\int \text{Qini}(k)\, dk - \text{baseline}}{\int \text{Qini}_{\text{perfect}}(k)\, dk - \text{baseline}}$$

Uplift@k — the observed treated-minus-control response-rate gap within the top-k% of customers by predicted score.

Both metrics carry a bootstrap 95% confidence interval ($n_{\text{bootstrap}} = 1000$, fixed `random_state` = 42 shared across every model within an arm, so CIs are directly comparable model-to-model).

4 Held-Out Results (test set, $n \approx 8{,}500$ per arm)

Refit on train+val ($n \approx$ 34,000 per arm), evaluated once on test — the only time test data is touched in the whole pipeline.

4.1 Men's E-Mail vs. No E-Mail

Every 95% CI includes zero: no model's ranking is statistically distinguishable from random targeting on the Men's arm.

4.2 Women's E-Mail vs. No E-Mail

X-Learner leads on point estimate but is statistically tied with S-Learner (near-identical, overlapping CIs) — both should be read as co-leading, not a clean single winner.

Model	Qini AUC	95% CI	Uplift@10%	Uplift@30%
S-Learner (GBM)	0.012	[-0.021, 0.048]	0.112	0.076
X-Learner (GBM)	-0.002	[-0.039, 0.033]	0.107	0.091
T-Learner (GBM)	-0.001	[-0.038, 0.033]	0.088	0.068
S-Learner (Logistic)	0.006	[-0.027, 0.037]	0.070	0.069

Model	Qini AUC	95% CI	Uplift@10%	Uplift@30%
X-Learner (GBM)	0.076	[0.042, 0.111]	0.079	0.076
S-Learner (GBM)	0.072	[0.037, 0.111]	0.068	0.069
T-Learner (GBM)	0.056	[0.020, 0.093]	0.080	0.067
S-Learner (Logistic)	0.017	[-0.015, 0.050]	0.065	0.064

Arm	Model-guided incr. visits	Random-targetin incr. visits	Treat-everyone incr. visits
Men's (S-Learner GBM)	1,458	1,437	4,789
Women's (X-Learner GBM)	1,454	931	3,103

4.3 Business-framed targeting impact (top 30% of customers, $n_{\textbf{targeted}} = 19{,}200$)

At a fixed 30% budget, model-guided targeting beats random targeting by a wide, practically meaningful margin for Women's (+56%); for Men's the two are statistically indistinguishable, consistent with Section 4.1's null result.

5 What Drives the Women's-Campaign Uplift (SHAP)

Scope: Women's E-Mail, X-Learner (GBM) only — the model with a Qini-AUC CI excluding zero. The Men's arm is excluded from this analysis entirely, not de-prioritized: every Men's model's CI includes zero, so its ranking is statistically indistinguishable from noise, and SHAP has no mechanism to flag that it may be attributing that same noise.

Because the X-Learner's blend weight $g(x)$ is a constant (the known randomization propensity), and both stage-2 regressors are tree ensembles, `shap.TreeExplainer` gives an exact (non-approximate) additive decomposition of the model's own predicted uplift — verified numerically (max reconstruction error $\approx 5.6 \times 10^{-17}$).

Rank	Feature	Mean |SHAP|	Direction
1	`womens` (prior women's-category purchase)	0.0222	Positive — prior purchasers show markedly higher predicted uplift
2	`recency` (months since last purchase)	0.0048	Positive — longer recency associated with higher predicted uplift
3	`history` ($ spent historically)	0.0040	Mixed / weaker — no consistent monotonic direction

Being an existing purchaser of women's merchandise is the dominant driver by a wide margin (mean |SHAP| $\approx 5\times$ the next feature). The highest-predicted-uplift test customer is a women's-category purchaser with 12 months' recency and $891 in historical spend; the lowest-predicted-uplift customer is a *men's-only* purchaser with 1 month's recency. **Targeting rule:** prioritize existing women's-category purchasers who have not bought very recently — consistent with, and explaining, the top-10–30%-by-predicted-uplift targeting shown to outperform random in Section 4.3.

6 Benchmarking Against Published Literature

Metric definition, verified: this project computes Qini AUC via `sklift.metrics.qini_auc_score`, confirmed from its source to be a **normalized** coefficient — actual-curve area minus the random-targeting baseline, divided by the same quantity for the theoretical perfect-targeting curve (the Radcliffe 2007 definition). It is a unitless ratio, not a raw area.

Metric definition, published sources: this normalization is **not confirmed** for the external figures below. Rößler & Schoder (2022) explicitly label their measure the "*unscaled*" Qini coefficient — the naming strongly implies it is **not** divided by the perfect-curve area, i.e. a different quantity in different units, which is why no numeric value from that paper is reproduced here. The Significance-First Splitting preprint (Hadjipantelis et al., 2026) reports a "Qini AUC" without citing scikit-uplift or stating the normalization explicitly (its listed dependencies are `rattus`, `econml`, and `scikit-learn`). A third, unrelated paper (UpliftBench; Singh, 2026) defines its own "Qini" as a raw, un-normalized integral of the gain curve — concrete confirmation that this scaling ambiguity is a real, active inconsistency in the uplift literature, not a hypothetical one.

Source	Method(s) / Dataset setup	Reported figure	Same normalization?
This project — Men's arm	S-/T-/X-Learner, Men's vs. No E-Mail	−0.002 to 0.012 (all CIs include 0)	Yes (verified — own code)
This project — Women's arm	S-/T-/X-Learner, Women's vs. No E-Mail	0.017 to 0.076 (X-/S-Learner CIs exclude 0)	Yes (verified — own code)
Significance-First Splitting (Hadjipantelis et al., 2026)	Tree/Forest/S-/T-Learner/Causal Forest, pooled "any e-mail" treatment	0.019 – 0.035	Not confirmed
Rößler & Schoder (2022), J. Interactive Marketing	15 uplift/HTE methods, Hillstrom Visit & Conversion as separate benchmarks	Not reproduced here	No — explicitly "unscaled"

Given that, the only claim this section makes with confidence is **qualitative**: every source surveyed — regardless of its exact Qini convention — describes Hillstrom as producing a *small, volatile* uplift signal relative to other benchmark datasets (e.g. Starbucks, Criteo), and finds that no single method dominates on it. That qualitative pattern is scale-independent — it holds whether a paper's numbers are normalized or not — and it is consistent with (not just asserted by) this project's own finding that the Men's arm is statistically null while the Women's arm is not. The specific *0.019–0.035* figures from Significance-First Splitting are shown for order-of-magnitude context only, **not** as a confirmed like-for-like comparison to this project's 0.012 (Men's) and 0.076 (Women's) — treat any apparent closeness or gap between those numbers as coincidental unless the normalization is independently confirmed.

7 Limitations

- **Sample size:** ∼8,500 rows per arm at test time is enough to separate the Women's leaders from random, but not enough to separate the Women's leaders from each other (X-Learner and S-Learner have overlapping CIs) or to say anything conclusive about the Men's arm.
- **Business-impact scaling:** `total_population` (≈64,000) is this experiment's own sample size, used as a stand-in for a real campaign's active customer file — it must be replaced with the actual file size before any dollar or headcount figure is treated as a production estimate.
- **No revenue framing:** results stay in incremental-visit terms; converting to $ requires an empirical conversion-given-visit rate and an average order value, both deployment-specific.
- **SHAP explains the model, not a verified mechanism:** it shows what the fitted X-Learner learned to associate with higher/lower predicted uplift on held-out data, not a causally validated behavioral explanation.

- **Extended comparison** (Decision Tree, Random Forest, MLP, TARNet, ensembles) was evaluated on the validation split only, not the held-out test set, and is not part of this report's headline figures.

8 Conclusion and Recommendation

Deploy a predicted-uplift targeting policy (top 10–30% by score) for the **Women's E-Mail** campaign; either the X-Learner or S-Learner (GBM) is defensible given their statistically tied performance. Do **not** deploy a targeting model for the **Men's E-Mail** campaign on current evidence — every candidate model's ranking is statistically indistinguishable from random at this sample size; more data or a richer feature set would be needed before revisiting it.

References

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